

attaching a white sticker suitably at the bottom of the laptop.

Calibrate the circuit before first use. Set preset VR1 at the centre and place the laptop in the docking station. Now turn VR1 slowly until IC2 goes to standby (no-oscillation) mode. Then

remove the laptop from the docking station, ensure that IC2 is enabled (pin 12 is low) and wait for the alarm sound. Repeat the process and adjust VR1 until you get the correct result. Note that the LED in the opto-sensor is permanently powered via resistor R2.

Similarly, you are free to experiment with the values of IC2 timing components C5, R3 and R4 for increasing or decreasing the delay time.

EFY note. During testing at EFY Lab, we used CX sensor from OMRON in place of CNY70. ●

1W LED FOR AUTOMOTIVE APPLICATIONS

■ T.A. BABU

This simple circuit lets you run a 1W LED from the battery of your car. IC MC34063 is used

here as a buck converter. It is a monolithic switching regulator sub-system intended for use as a DC-DC converter. The device consists of an internal temperature-compensated reference,

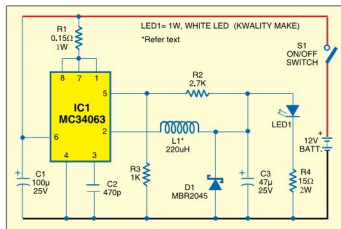


Fig. 1: 1W white LED for automotive

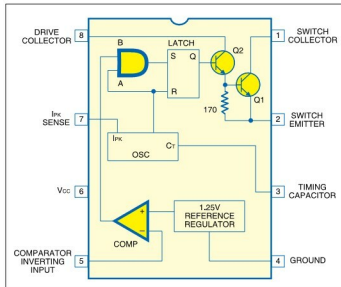


Fig. 2: Internal diagram of MC34063

a comparator, a controlled duty-cycle oscillator with an active current-limit circuit, a driver and a high-current output switch. These functions are contained in an 8-pin dual in-line package. Another major advantage of the switching regulator is that it allows increased application flexibility of the output voltage.

Fig. 1 shows the circuit of 1W LED for automotive applications. Fig. 2 shows the internal diagram of IC MC34063, while Fig. 3 shows a 1W LED. The non-linear voltage-to-current re-

lationship and variation in forward voltage with temperature necessitate the use of a 320mA constant-current power source as provided by this supply. The current through LED1 is sensed by a 15-ohm resistor (R4) and the sensed voltage is fed to the controller to maintain a constant current. Resistor R1 limits the peak current given to IC1. With capacitor C2 (470pF), the oscillator runs at about 100 kHz.

Once the circuit is ready, do not connect the LED directly: First, use a 10-ohm, 5W resistor as the dummy load and verify the current and voltage across the dummy. Run the dummy for a few minutes. If the result is satisfactory, replace the dummy with the actual LED1. Inductor L1 used here is a salvaged drum core of size 8×10mm² with 70 turns of 33SWG enameled copper wire, giving an inductance of about 220 μH.

Assemble the circuit on a general-purpose PCB and enclose in a suitably small cabinet. Solder the IC base for easy troubleshooting. Fix the



Fig. 1: 1W LED

inductor and the LED1 (using a suitable heat-sink) in the PCB. Like all light bulbs, LED lighting is not waterproof. If used